Task 2

2.1 Company Mission, Core Values, and Corporate Background

The virtual company designed for this study is Hong Kong Compliance Tech (HKCT), a cross-jurisdictional RegTech firm based in Hong Kong, dedicated to serving financial institutions in the Asia-Pacific region. HKCT is committed to solving compliance pain points for multinational financial institutions through technological innovation, helping the financial industry achieve a compliant, fair, transparent, and efficient digital transformation (Hong Kong Compliance Tech, 2025).

(1) Company Mission: Empower compliance with technology, protect rights with fairness, and provide algorithmic fairness and fair lending compliance solutions tailored to major global jurisdictions for Hong Kong and multinational financial institutions. HKCT aims to reduce cross-jurisdictional compliance costs, identify hidden algorithmic discrimination risks, safeguard the legitimate rights and interests of financial consumers, and promote the healthy and sustainable development of the financial industry (Hong Kong Compliance Tech, 2025).

(2) Core Values: HKCT adheres to four core values throughout product design, technology development, and service delivery:

- Compliance Authenticity: Do not whitewash, conceal, or circumvent real risks. The primary goal of tool design is to identify genuine compliance issues, rather than merely meeting minimum compliance documentation requirements (Hong Kong Compliance Tech, 2025).
- Algorithmic Fairness: Uphold technological neutrality, prioritize the reduction of algorithmic discrimination, ensure group fairness, and favor neither financial institutions nor consumers, basing decisions on data and facts (Consumer Financial Protection Bureau [CFPB], 2025; Hong Kong Monetary Authority [HKMA], 2025).
- Transparency and Traceability: The entire tool operation process, algorithmic decision-making logic, and compliance determination results shall be recordable, auditable, and traceable, meeting the transparency requirements of supervisors and clients (HKMA, 2025; Office of the Comptroller of the Currency [OCC], 2026).

- Customer Value First: Deeply understand the actual business scenarios and compliance pain points of clients (financial institutions). Product design balances compliance, practicality, and cost-effectiveness, helping clients create long-term value (Hong Kong Compliance Tech, 2025).

(2) Corporate Background and Development Stage: Hong Kong Compliance Tech was established in 2023, with its headquarters located at the Hong Kong Science and Technology Parks. It is a technology enterprise transitioning from startup to growth stage. The team has a total of 25 members, with core members comprising experienced Hong Kong financial compliance experts, AI algorithm engineers, data analysts, and product managers. Team members have an average of over 8 years of experience in financial compliance or FinTech, with deep expertise in financial regulatory rules across core Asia-Pacific markets such as Hong Kong, the United States, and Singapore, and proficiency in algorithmic fairness techniques, rule engine development, and compliance automation solutions (Hong Kong Compliance Tech, 2025). The company focuses on three core product lines: cross-jurisdictional algorithmic compliance, fair lending detection, and AI governance tools. It has already served three local Hong Kong banks (including The Bank of East Asia) and two multinational financial institutions, receiving positive client feedback (BEA, 2024; Hong Kong Compliance Tech, 2025).

(4) Business Goals and Development Plan: The company has established a clear three-year development plan: within one year, complete core client coverage in the Hong Kong market and enhance the U.S.-Hong Kong rule adaptation functions; within two years, expand into the Singapore and Australian markets, with product functionality covering full-dimension algorithmic bias detection; within three years, become a leading cross-jurisdictional algorithmic compliance service provider in the Asia-Pacific region, covering four core markets (Hong Kong, the United States, Singapore, and Australia), with an annual revenue target of HKD 12 million, a client retention rate of no less than 85%, and gradually build an Asia-Pacific financial compliance technology ecosystem (Hong Kong Compliance Tech, 2025).

2.2 Tool Service Objects and Stakeholder Conflict Analysis

(1) Core Service Objects

The FairLink tool designed in this study serves a dual-layer structure of "direct paying clients + indirect beneficiaries," covering three core groups: financial institutions, regulatory authorities, and financial consumers.

- Direct Paying Clients: Compliance departments, risk management departments, and credit approval departments of The Bank of East Asia and similar multinational financial institutions. These clients are the purchasers and direct users of the tool, with core needs to reduce compliance costs, improve audit efficiency, avoid regulatory penalties, and ensure normal business operations (BEA, 2024; Hong Kong Compliance Tech, 2025).
- Indirect Beneficiaries: These include regulatory authorities, such as the Hong Kong Monetary Authority (HKMA), the U.S. Consumer Financial Protection Bureau (CFPB), and the Office of the Comptroller of the Currency (OCC). The tool can provide automated compliance reports, algorithmic fairness data, and early risk warning information, assisting supervisors in conducting efficient oversight and precise enforcement (CFPB, 2025; HKMA, 2025; OCC, 2026). They also include financial consumers, such as individual credit applicants of The Bank of East Asia (both Hong Kong local customers and U.S. market customers). By identifying and reducing algorithmic discrimination, the tool helps ensure that consumers receive fair, impartial, and transparent credit approval services, thereby protecting their legitimate rights and interests (CFPB, 2025; HKMA, n.d.).

(2) Core Conflicts of Interest

Different service objects have inherently different core demands, giving rise to three core conflicts of interest that must be balanced in tool design:

- Financial Institutions vs. Regulatory Authorities: Financial institutions seek the lowest compliance costs, highest approval efficiency, manageable risks, and minimal penalty exposure. They prefer tools with flexible rules, low false-positive rates, and convenient operation. Regulatory authorities pursue strict compliance standards, algorithmic fairness, transparency, auditability, and zero tolerance for risk. They prefer tools with rigorous rules, accurate detection, complete records, and timely warnings. These two sets of demands are clearly in tension (CFPB, 2026; HKMA, 2025).

- Financial Institutions vs. Financial Consumers: Financial institutions pursue profit maximization, risk minimization, approval efficiency priority, and low complaint handling costs. They may tend to simplify approvals, increase rejection rates for high-risk customers, and reduce explanation costs. Financial consumers pursue fair approval, reasonable approval rates, clear reasons for denial, and protected rights. They expect the tool to strictly identify discrimination, protect vulnerable groups, and provide detailed explanations. These core interests are potentially conflicting (CFPB, 2025; HKMA, n.d.).
- Hong Kong Market vs. U.S. Market: The Hong Kong business unit of The Bank of East Asia expects the tool to adapt to Hong Kong's more flexible rules and reduce compliance burdens. The U.S. business unit must strictly comply with stringent U.S. rules to avoid penalty risks. The differing rule requirements across departments further complicate tool design (CFPB, 2026; HKMA, 2025).

(3) Design Rationale

The design of the FairLink tool prioritizes compliance authenticity, technical practicality, and core client demands. The key rationales include:

- Market Demand Priority: Multinational financial institutions have prominent cross-jurisdictional compliance pain points, strong willingness to pay, large market size, and high growth potential. The product has clear commercial value (BEA, 2024; Hong Kong Compliance Tech, 2025).
- Technical Capability Match: HKCT's compliance technology team has deep expertise in algorithmic fairness techniques and rule engine development, with mature technology and rich experience, enabling efficient cross-jurisdictional rule adaptation and automated detection functions (Hong Kong Compliance Tech, 2025).
- Risk Prevention Priority: The tool is designed to identify genuine algorithmic discrimination risks and safeguard compliance baselines, rather than simply catering to financial institutions' demands for low costs. This ensures that the tool balances the

interests of all parties while adhering to the core principles of compliance and fairness (CFPB, 2025; HKMA, 2025).

2.3 Tool Positioning: Identifying Genuine Risks Rather Than Documentary Compliance

(1) Core Tool Positioning

The core positioning of the FairLink tool is to prioritize the identification of genuine algorithmic discrimination risks and achieve proactive risk prevention, rather than merely meeting minimum compliance documentation requirements. The design philosophy is that the essence of compliance is to eliminate actual risks and ensure fairness and justice, not simply to generate compliance reports or record approval data. Tools that only satisfy documentary compliance cannot truly identify hidden algorithmic biases, are insufficient to prevent systemic discrimination risks, and cannot help financial institutions fundamentally resolve cross-jurisdictional compliance issues (CFPB, 2025; HKMA, 2025). Therefore, the FairLink tool consistently takes risk identification, risk warning, and risk remediation as its core objectives, while also providing compliance documentation generation functions, achieving the dual value of "risk prevention as the primary focus, documentary compliance as a secondary support" (Hong Kong Compliance Tech, 2025).

(2) Key Design Choice (Reflecting the Risk-Priority Positioning)

To implement the above positioning, one core design choice has been made that is not mandated by regulation but is necessary for risk prevention: mandatory inclusion of a cross-group approval difference quantitative detection module.

- Module Function: This module automatically calculates approval rate differences, denial rate differences, credit score distribution differences, and other key metrics for core group characteristics such as gender, age, income, and geographic region, enabling real-time identification of systemic unfairness between groups (Hong Kong Compliance Tech, 2025).
- Non-Mandatory Nature: Neither U.S. nor Hong Kong regulatory rules mandate that financial institutions perform routine quantitative difference detection for all group characteristics; only regular audits for core protected characteristics (e.g., gender, race)

are required (CFPB, 2026; HKMA, 2025).

- Risk Prevention Value: This module can capture hidden biases (e.g., age discrimination, geographic discrimination) that are not yet mandated by regulation but exist in practice, providing early warnings and prompting remediation before algorithmic discrimination triggers regulatory attention or consumer complaints. This fundamentally reduces compliance and reputational risks, truly achieving proactive risk prevention (CFPB, 2025; HKMA, 2025).

2.4 Responsibility Allocation and Impact Analysis of Tool Errors

FairLink tool, as a technical assistance tool, is not absolutely perfect and may exhibit four typical types of errors: false positives, false negatives, model performance drift, and incorrect jurisdiction rule configuration. The responsible parties, specific harms, and affected core stakeholders differ clearly for each error type, as analyzed below:

- False Positive (Compliant Misjudged as Non-Compliant): refers to the tool incorrectly determining that an approval outcome without algorithmic discrimination risk is non-compliant. Responsibility: The Bank of East Asia bears primary responsibility. Specific Harm: reduced approval efficiency for the bank, extended compliance review processes, increased operational costs, and risk of losing high-quality customers. Core Affected Parties: The Bank of East Asia (business department, risk control department) (BEA, 2024; Hong Kong Compliance Tech, 2025).
- False Negative (Non-Compliant Misjudged as Compliant): refers to the tool incorrectly determining that an approval outcome with algorithmic discrimination risk is compliant. Responsibility: The Bank of East Asia bears primary responsibility; the tool developer bears secondary responsibility. Specific Harm: financial consumers are unfairly denied credit, miss out on credit opportunities, suffer economic loss, and have their rights and interests harmed. Core Affected Parties: financial consumers (credit applicants) (CFPB, 2025; HKMA, n.d.).
- Model Performance Drift (Decline in Algorithmic Fairness Detection Accuracy): refers to the gradual decline in the accuracy of the tool's built-in algorithmic fairness detection model as market conditions, customer characteristics, and data distributions change,

rendering it unable to accurately identify algorithmic bias. Responsibility: The Bank of East Asia bears primary responsibility; the tool developer bears maintenance and update responsibility. Specific Harm: hidden algorithmic discrimination persists, compliance risks accumulate, the risk of regulatory penalties increases, and the bank's reputation is damaged. Core Affected Parties: The Bank of East Asia, financial consumers, regulatory authorities (CFPB, 2025; OCC, 2026).

- Incorrect Jurisdiction Rule Configuration (U.S./Hong Kong Rule Switching Error): refers to human operational error or system failure causing the tool to incorrectly load the regulatory rules of the wrong jurisdiction (U.S. or Hong Kong), rendering its compliance determination standards invalid. Responsibility: The Bank of East Asia bears primary responsibility. Specific Harm: directly triggers cross-jurisdictional compliance violations, leading to regulatory fines, business restrictions, and mandatory compliance remediation. Core Affected Parties: The Bank of East Asia, regulatory authorities (CFPB, 2026; HKMA, 2025).

Above all, financial consumers bear the most direct and most severe actual harm from tool errors (unfair credit denial, economic loss, rights infringement). The Bank of East Asia bears the primary economic costs, compliance risks, and reputational losses. Regulatory authorities bear the indirect impact of wasted regulatory resources and hindered achievement of supervisory objectives. Throughout the tool design process, various error scenarios have been fully considered, and measures such as multi-layer verification mechanisms, manual review steps, regular model updates, and operational log retention have been implemented to minimize the probability of errors and reduce the resulting harm (Hong Kong Compliance Tech, 2025).

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